

Summary

The radio frequency band spanning from 7.125 GHz to 24 GHz, referred to as Frequency Range 3 (FR3), holds significant promise for the evolution of next-generation cellular systems. This project aims to extend the current COSMOS testbed at WINLAB to facilitate experimental studies of coexistence between terrestrial and satellite systems using software-defined radios (SDR) to emulate dense 5G cellular networks and satellite equipment. Specifically, the project focuses on spectrum sharing between terrestrial 5G and active commercial satellites in the 12.2-12.7 GHz band, and adjacent band coexistence between terrestrial 5G, operating at 10-10.5 GHz, and passive scientific satellites in the 10.6-10.7 GHz band. Metamaterial software-defined beamforming is used to emulate coexistence with satellites like non-geostationary orbit fixed satellite service (NGSO-FSS), and hot-cold calibration and comparison to datasets from on-orbit AMSR-E and AMSR2 sensors is used to enhance the reliability and realism of radiometer interference experiments. The project unfolds in three key thrusts:

- (1) Designing, validating, and deploying FR3 SDR-based heterodyne devices, emulating 5G New Radio (NR) and SAT waveforms, and developing coexistence emulation methods.
- (2) Employing emulation experiments to craft centralized Machine Learning algorithms for integrated radio resource management in sharing between terrestrial 5G and active commercial satellites.
- (3) Leveraging emulation experiments to evaluate radiometer sensitivity to terrestrial 5G interference and develop Machine Learning algorithms for interference identification and mitigation by passive scientific satellites.

One outcome from thrusts 2 and 3 is the measurement of the fraction of spectrum allocated to terrestrial 5G cellular systems that is lost due to the strategies used to mitigate interference to satellites.

Research Progress

SDR-based Heterodyne Transceiver for FR3 spectrum using COSMOS Sandbox

- ❑ We built and tested a working FR3 capable SDR heterodyne transceiver up to 24 GHz (Figure 4)
- ❑ Experimental validation of frequency conversion up to 24 GHz completed
- ❑ Optimized filtering/amplifier chain enables clean up-conversion and realistic FR3 coexistence emulation (5G NR at 10–10.5 & 12.2–12.7 GHz, DBS/NGSO FSS at 12.2–12.7 GHz) in COSMOS Sandbox
- ❑ Next Steps:
 - Architecture finalized for FR3-band PCB design. Fabrication planned for 2025 with integrated up/down conversion up to 24 GHz

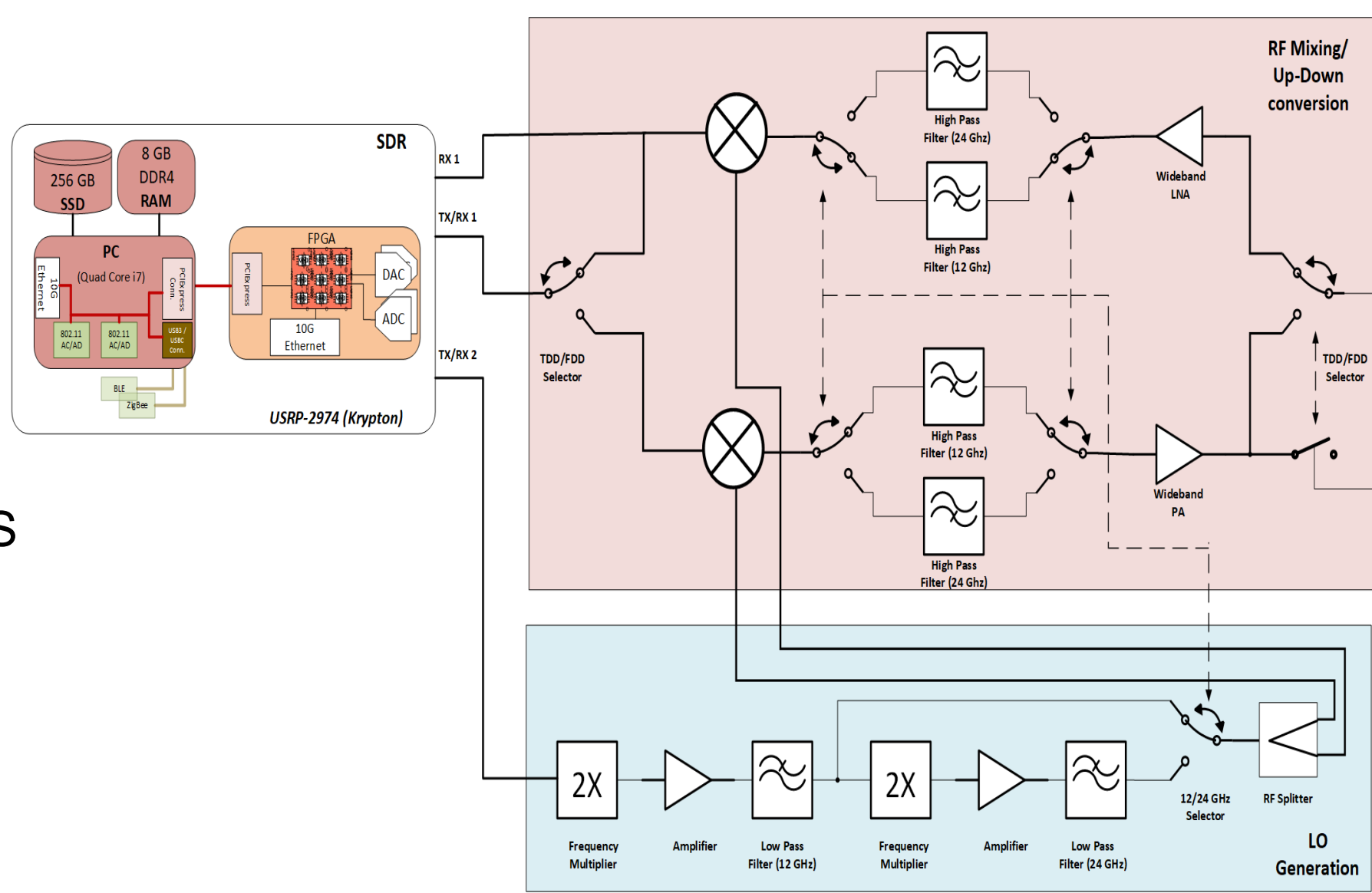


Figure 1: Proposed SDR-Based Heterodyne Transceiver for FR3 Tested for full-band operation up to 24 GHz

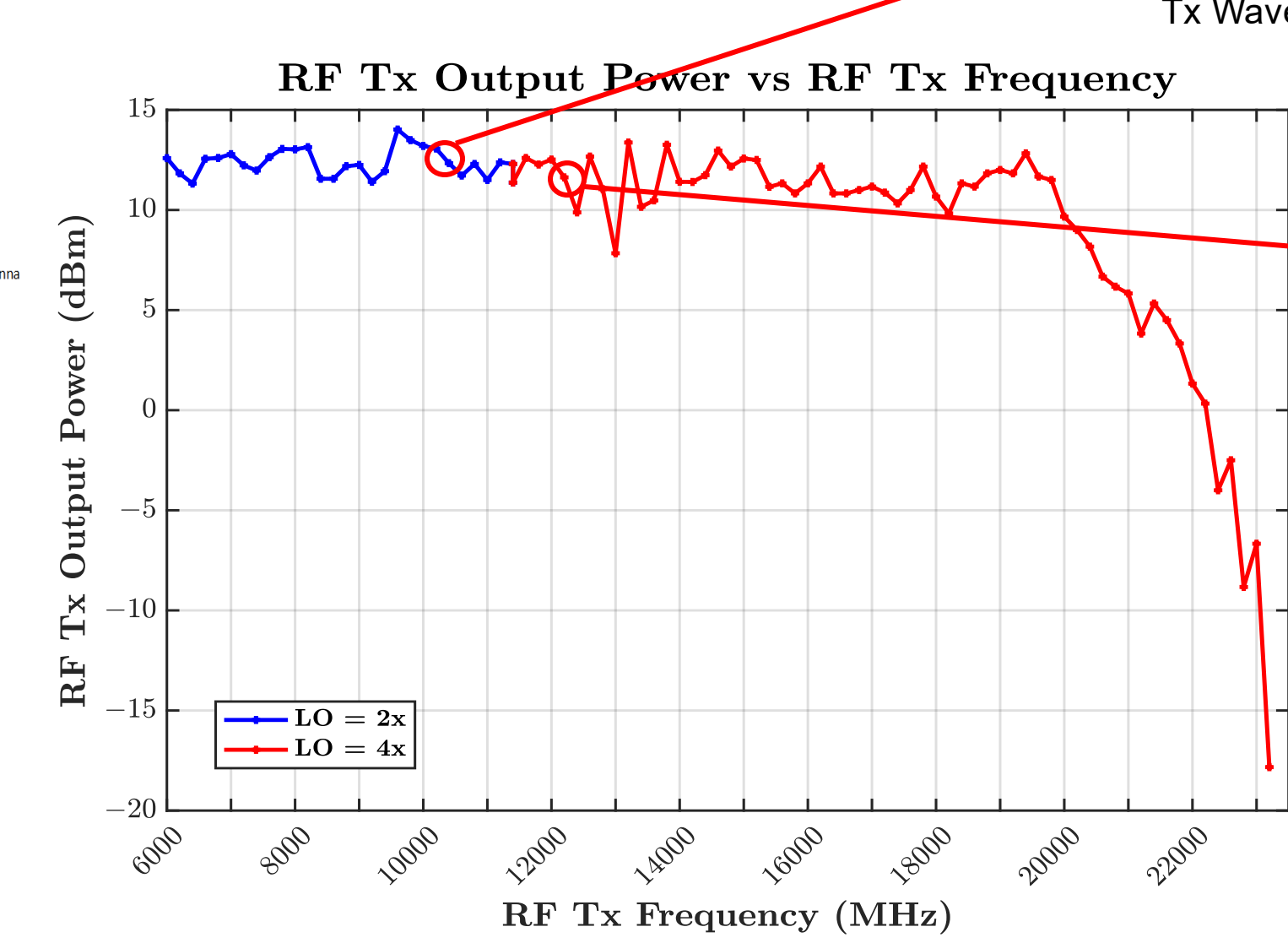


Figure 2: FR3 power levels (6–23 GHz). Roll-off > 20 GHz due to PA bandwidth; key bands: 10–10.5 & 12.2–12.7 GHz

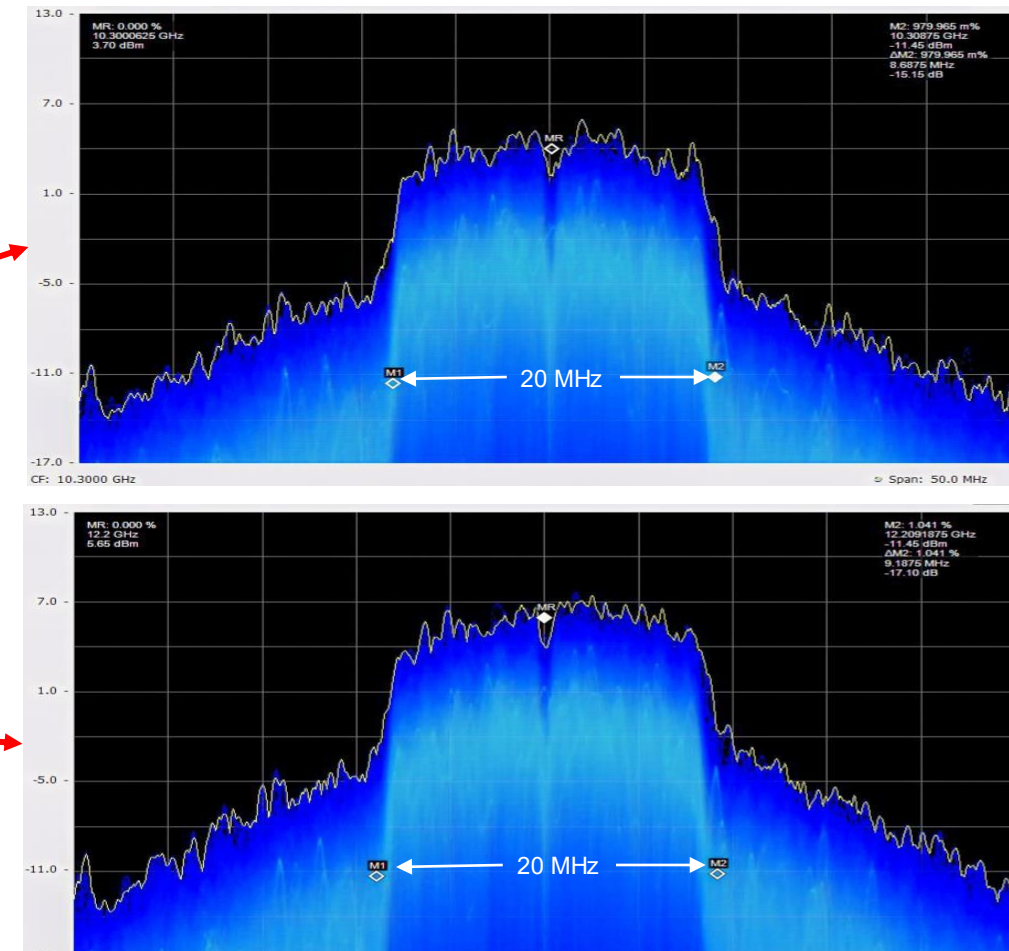


Figure 3: Upconverted OFDM Spectrum at (a) 10.3 (top) & (b) 12.2 GHz (bottom)

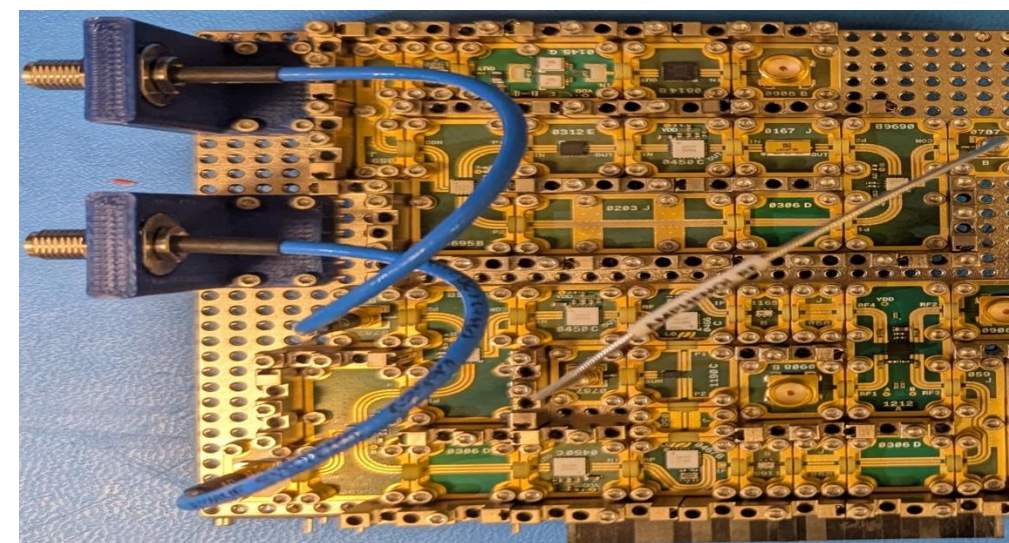


Figure 4: FR3 Heterodyne Transceiver (Quantic X-Microwave Prototyping Board)

Emulation of 5G NR Protocol Stack and DBS SAT Downlink Signals in 12 GHz

- ❑ Spectrum Management - Digital Broadcasting Satellite Emulation
 - Digital Video Broadcasting Satellite Generation 2 (DVB-S2) Transmitter/Receiver through GNU Radio w/ SDR B210 & X310
 - Simulate 64-bin & 256 QAM 5G OFDM through Amarisoft
 - Baseband Spectrum Management through Linear Quadratic Regulator Approach; preserved DVB-S2 link 82-99%
- ❑ Next Steps:
 - Integrate 12 GHz FR3 upconverter with current SDR & GNU Radio setup
 - Develop Reinforcement Learning (RL) control model for DVB-S2 & 5G systems coexistence in 12.2-12.7 GHz

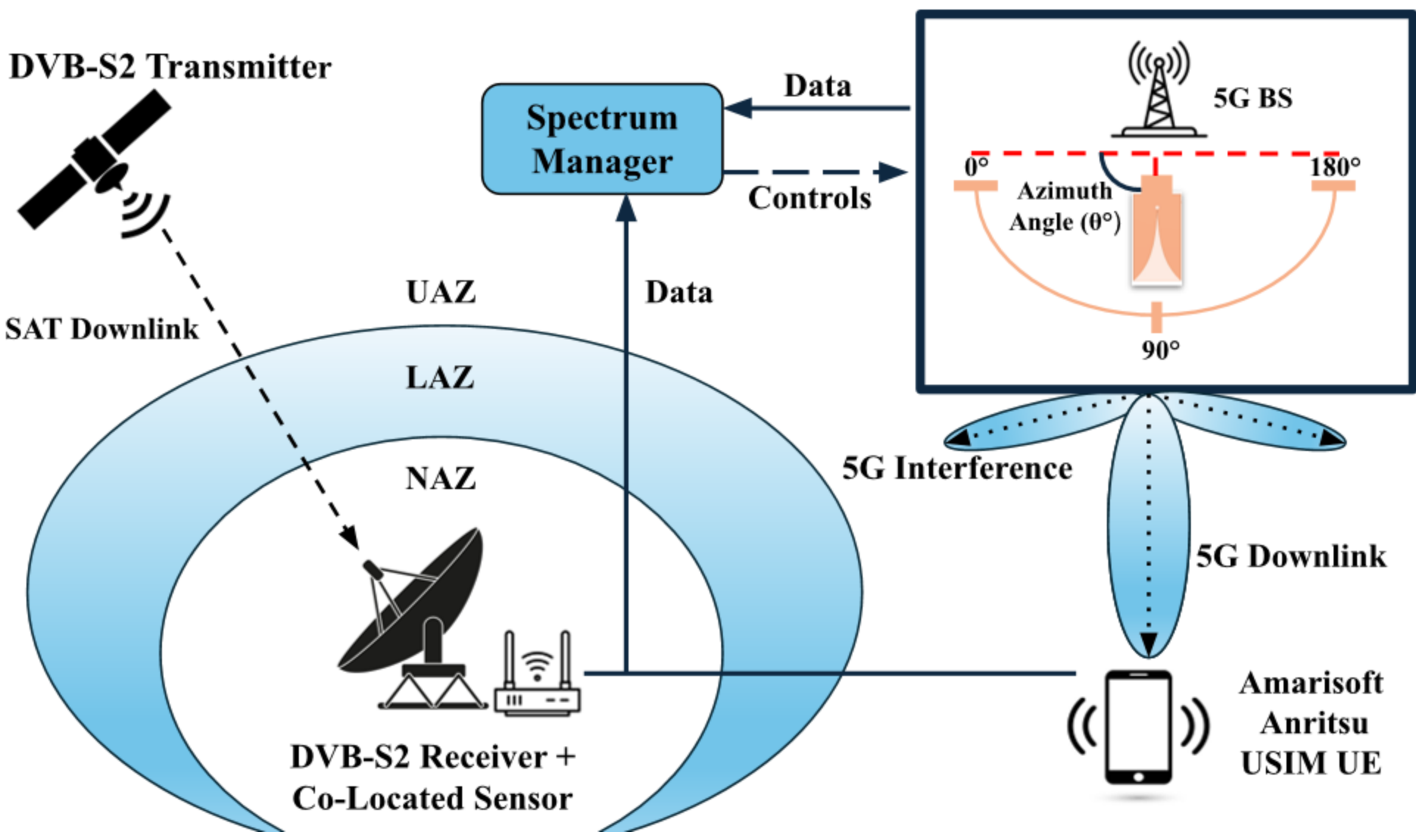


Figure 5: Spectrum Management Framework : with 5G BS beam sweeping 0°–180° azimuth and satellite interference control

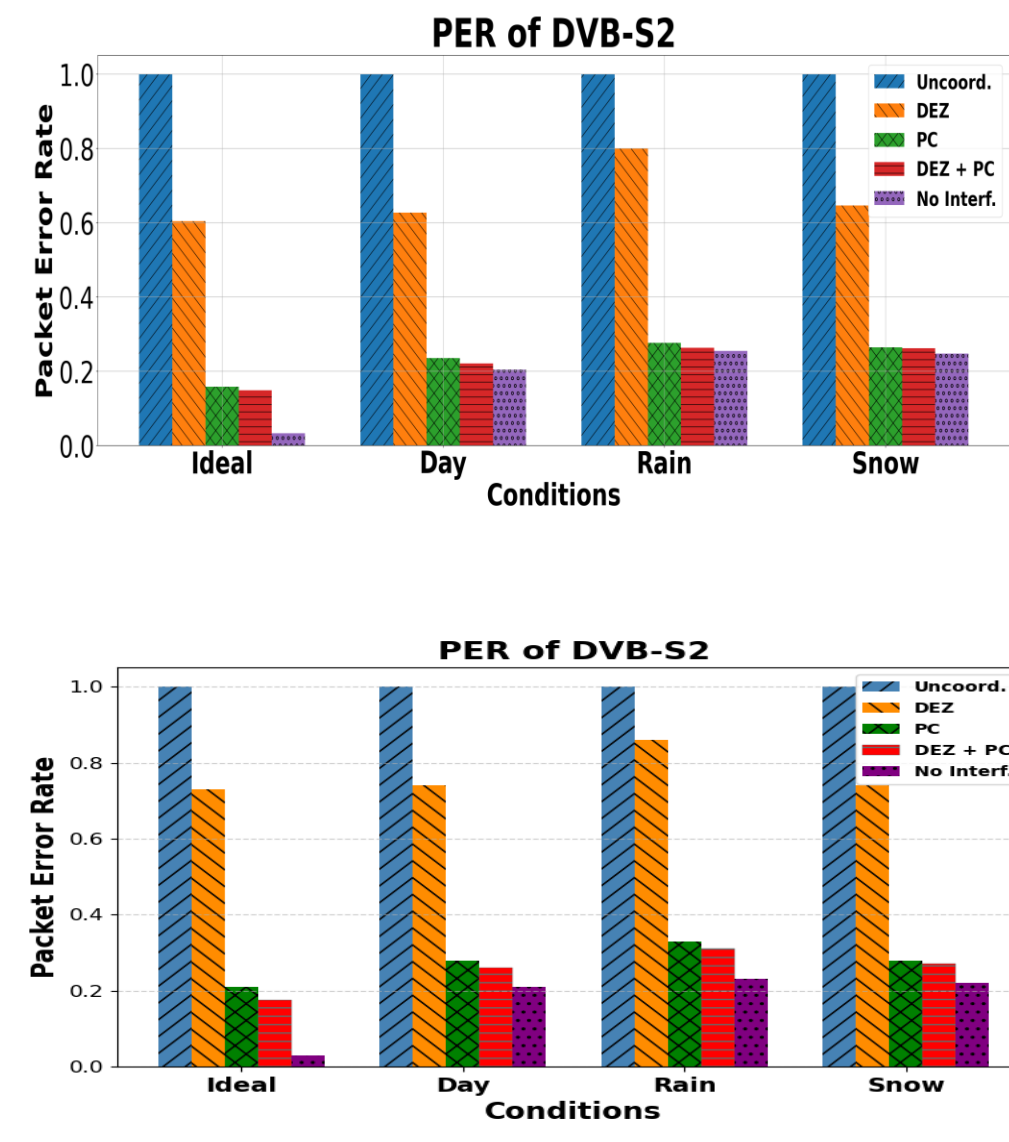


Figure 6: PER of DVB-S2 with (a) Single (top) and (b) Multiple omnidirectional 5G BS (bottom)

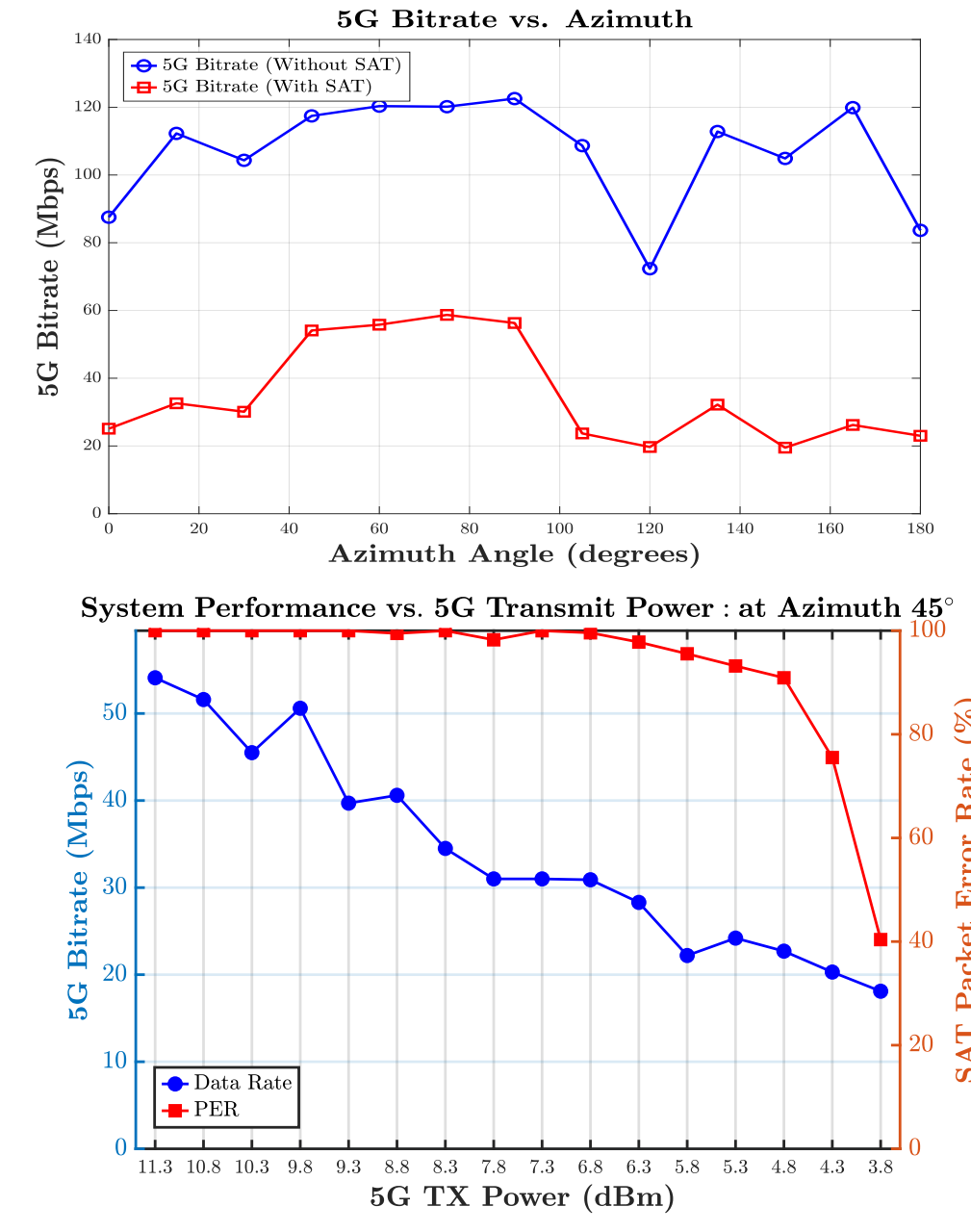


Figure 7: (a) 5G Bitrate vs. Azimuth Angle during SAT Scenarios w/ Constant Power (top) (b) 5G Bitrate and SAT PER vs. TX Power w/ Constant Azimuth (bottom)

Modeling 5G–Satellite Coexistence at 10–10.7 GHz

- ❑ Future 5G deployment at 10–10.5 GHz is estimated by mapping Gompertz-based population forecasts to network demand and base station density
- ❑ Radio Frequency Interference (RFI) to Earth exploration satellites in the adjacent 10.6–10.7 GHz band is evaluated using array-aware propagation modeling with beam tilt and angular loss
- ❑ Our case study of New York City reveals how spectral efficiency, elevation angle, and sidelobes jointly shape coexistence margins and deployment scale
- ❑ Next Steps:
 - Integrate real-world base station angle and user equipment into base station planning and beamforming scenarios
 - Validate array–sensor interference with over-the-air SDR experiments and fabricated antenna prototype

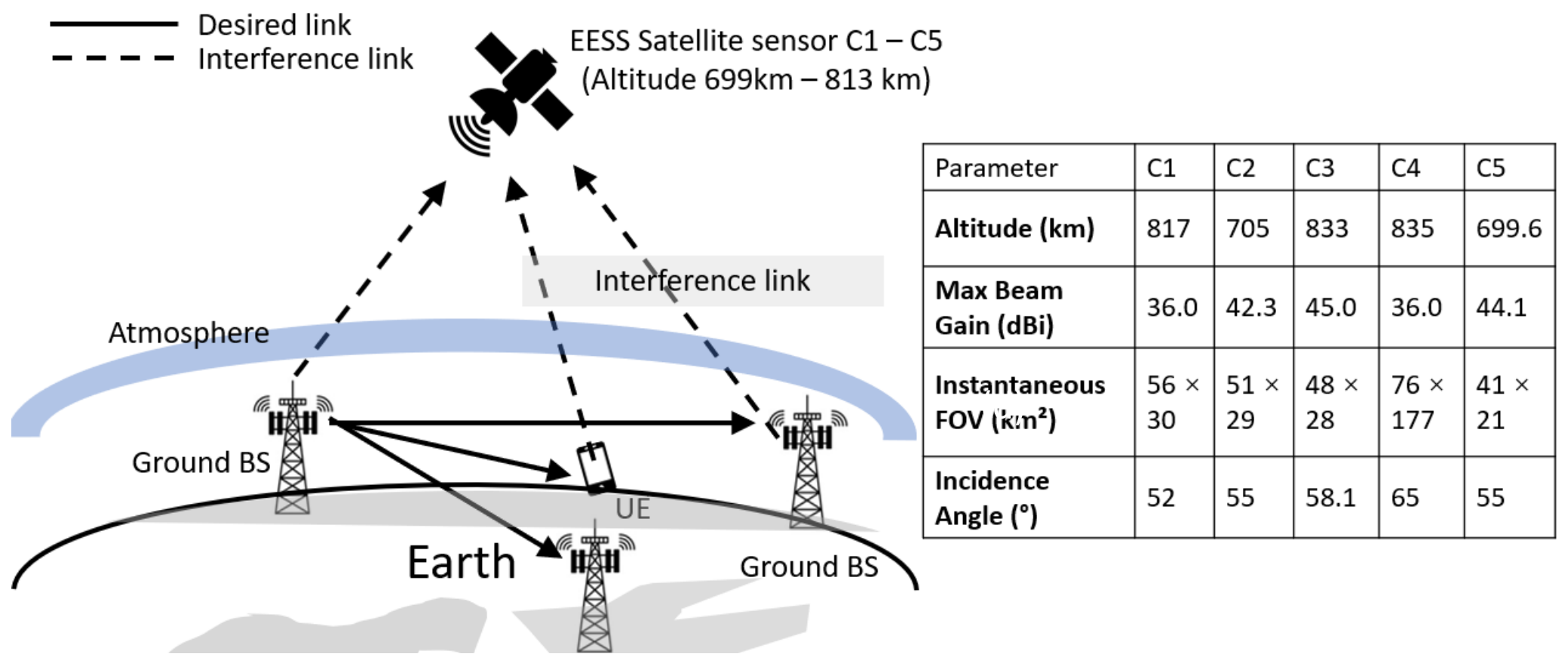


Figure 8: 5G–Satellite Coexistence Framework and the parameters of the EESS sensors

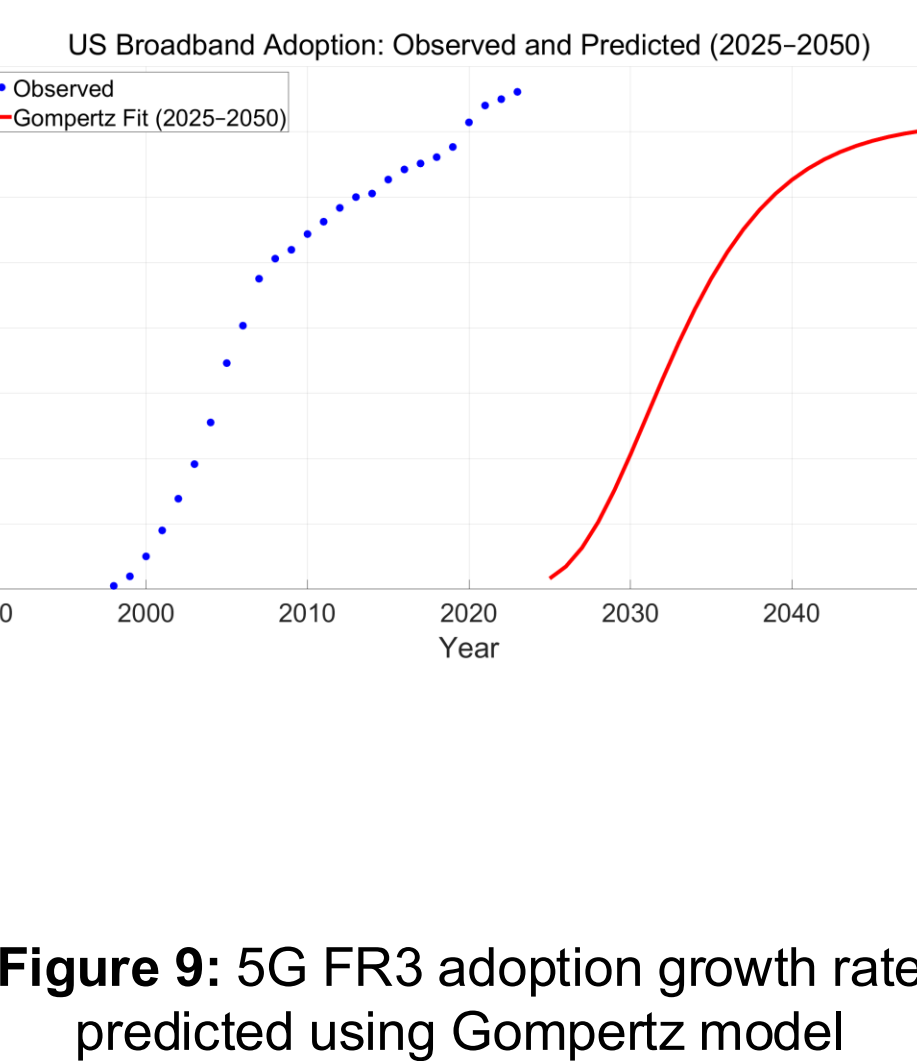


Figure 9: 5G FR3 adoption growth rate predicted using Gompertz model

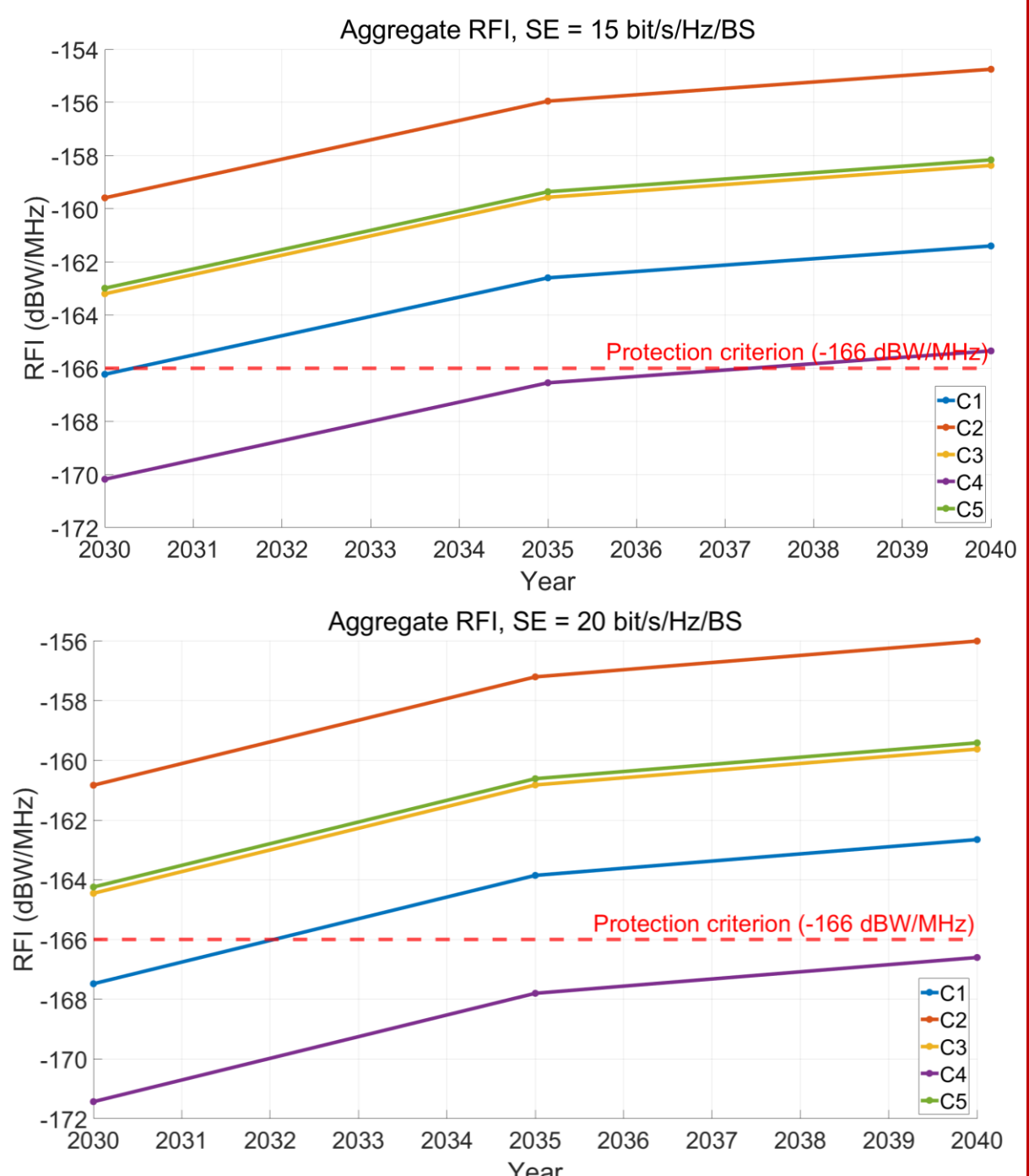


Figure 10: Simulated result of aggregate Radio Frequency Interference (RFI) when Spectral Efficiency = 15 (top) and 20 (bottom) bits/s/Hz/BS

Spectrum Coexistence of 5G Terrestrial Networks with NGSO-FSS and Passive Weather Sensors

- ❑ COSMOS Sandbox Emulation of Spectrum Coexistence between 5G Terrestrial Networks and Passive Sensors on Earth Observation Satellites in 10.6–10.7 GHz
 - Data-Driven Approach to Modeling Sensor Sensitivity and Identifying RFI Sources
 - Dynamic Spectrum Access for Proactive RFI Mitigation
- ❑ Spectrum Coexistence between 5G Terrestrial Networks and 12.2-12.7 GHz NGSO-FSS
 - Emulation of Starlink-like NGSO-FSS Signals Incorporating Full Frame Structure
 - Radio Resource Management Based on Interference-to-Noise Ratio (INR) Feedback

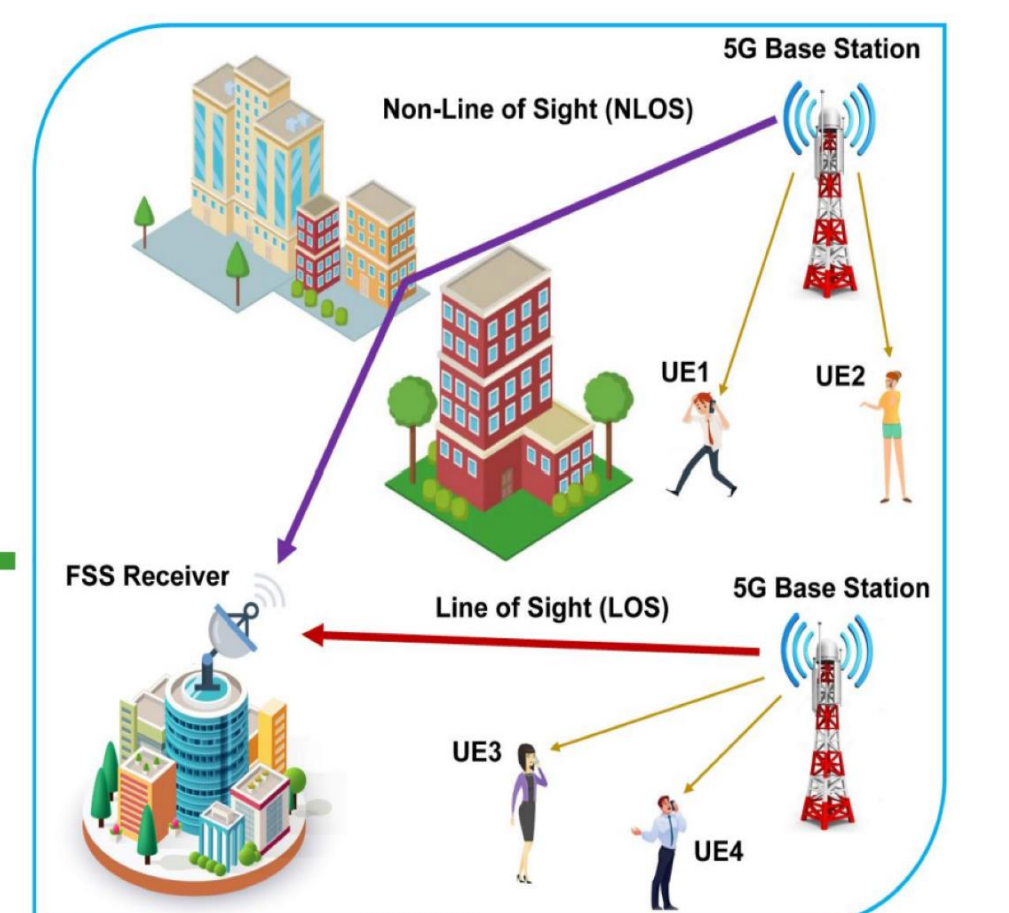
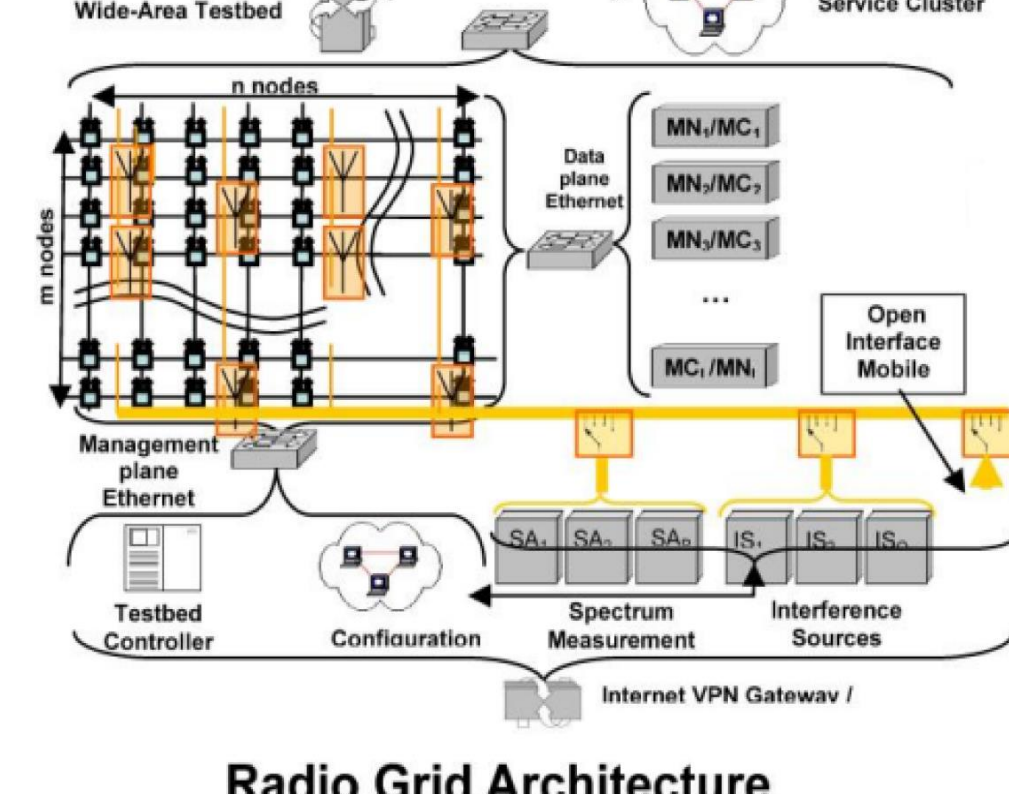


Figure 11: COSMOS GRID@WINLAB with Control Plane, Measurement Infrastructure and Radio Mapping

References

[1]. FCC Technical Advisory Council, "A Preliminary View of Spectrum Bands in the 7.125 - 24 GHz Range; and a Summary of Spectrum Sharing Frameworks," August 2023. [Online].
[2]. J. Chen, D. Zhou, L. Zhong, and D. Li, "Coexistence study of 5G in 10–10.5 GHz and passive Earth exploration-satellite service in 10.6–10.7 GHz," Proc. Int. Wireless Communications and Mobile Computing Conf. (IWCMC), Marrakesh, Morocco, 2023, pp. 711–716, doi: 10.1109/IWCMC58020.2023.10182698.
[3]. S. Mandava, R. B. Rashid, S. Vosoughitabar, C.-T. M. Wu, I. Seskar, and N. B. Mandayam, "Software defined radio based emulation of sat–terrestrial network coexistence," in Proc. IEEE Global Communications Conference (GLOBECOM), 2025, accepted for publication.
[4]. S. Mandava, R. B. Rashid, D. Gao, V. Malaviya, P. Maddala, C.-T. M. Wu, I. Seskar, and N. B. Mandayam, "SAT–terrestrial coexistence with adaptive beam and power control: An SDR-based emulation," submitted to IEEE Future Networks World Forum (FNWF), 2025.
[5]. D. Gao, R. B. Rashid, V. Malaviya, C.-T. M. Wu, I. Seskar, and N. B. Mandayam, "Emulation-Based Coexistence Analysis of Future 5G FR3 Networks and Passive Earth Exploration-Satellite Sensors in the 10–10.7 GHz Band," in preparation.

References